Lab Assignment 5

In [103…

```
import numpy as np # Step 1: Import libraries
import pandas as pd
```

In [104…

```
df = pd.read_csv("Social_network_dataset.csv") # Step 2: Load dataset
df
```

Out[104…

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

In [105…

```
df.isnull().sum() # Step 2: Load dataset
```

Out[105…

```
User ID            0
Gender             0
Age                0
EstimatedSalary    0
Purchased          0
dtype: int64
```

In [106…

```
from sklearn.preprocessing import LabelEncoder # Step 4: Encode categorical data
le = LabelEncoder()
df["Gender_encoded"] = le.fit_transform(df["Gender"])
df = df.drop("Gender", axis=1)
df.head()
```

Out[106…

	User ID	Age	EstimatedSalary	Purchased	Gender_encoded
0	15624510	19	19000	0	1
1	15810944	35	20000	0	1
2	15668575	26	43000	0	0
3	15603246	27	57000	0	0
4	15804002	19	76000	0	1

In [107…

```
X = df.drop("Purchased", axis=1) # Step 5: Define features (X) and target (y)
y = df["Purchased"]
```

In [108…

```
from sklearn.model_selection import train_test_split # Step 6: Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=0
)
```

In [109…

```
from sklearn.preprocessing import StandardScaler # Step 7: Feature Scaling (IMPO
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

In [110…

```
from sklearn.linear_model import LogisticRegression # Step 8: Train model
model = LogisticRegression()
model.fit(X_train, y_train)
```

Out[110…

```
▾ LogisticRegression
▶ Parameters
```

In [111…

```
y_pred = model.predict(X_test) # Step 9: Predict
print("Predicted values:\n", y_pred)
```

```
Predicted values:
 [0 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0 0 1 0 0 1 0 1 0 1 0 0 0 0 0 0 1 0 0 0 0
 0 0 1 0 0 0 0 1 0 0 1 0 1 1 0 0 0 1 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 1 0 0 0
 0 0 0 0 1 1 1 1 0 0 1 1 0 1 0 0 0 1 0 0 0 0 0 0 1 1]
```

In [112…

```
from sklearn.metrics import confusion_matrix # Step 10: Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)
```

```
Confusion Matrix:
 [[66  2]
 [ 8 24]]
```

In [113…

```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_sc
```

In [114…

```
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))
print("F1 Score:", f1_score(y_test, y_pred))
```

```
Accuracy: 0.9
Precision: 0.9230769230769231
Recall: 0.75
F1 Score: 0.8275862068965517
```